QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION, OCTOBER 2026
FIFTH SEMESTER
DSE –MG5DSEECO302

ACTUARIAL SCIENCE AND RISK MANAGEMENT IN INSURANCE

(2024 ADMISSION ONWARDS)

Duration2 hrs **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A- Multiple Choice Questions

Answer ***all*** questions

1. Which element of a life insurance portfolio represents a statistical table showing the probability of a person dying before their next birthday at any given age? [CO1]
   A) Solvency Table
   B) Claim Development Matrix
   C) Mortality Table
   D) Underwriting Schedule
2. The Actuarial valuation technique that assesses whether an insurance firm has sufficient assets to cover future claims and policy obligations is known as the verification of: [CO1]

   A) Market Elasticity
   B) Solvency Margin
   C) Moral Hazard
   D) Demographics
3. If the price elasticity of demand for a specific auto insurance product is highly inelastic, how will consumers respond to a moderate increase in premium rates**?** [CO2]

   A) They will drastically reduce their policy coverage amounts.
   B) They will completely stop buying the coverage.
   C) Their consumption volume will change very insignificantly.
   D) They will instantly substitute it with bancassurance products.
4. Which micro-insurance distribution framework involves a bank selling insurance policies to its existing banking customer base? [CO2]
   A) GSLI Network
   B) Group Superannuation
   C) Bancassurance
   D) Mutual Pool Account
5. According to the foundational principles of behavioural economics in insurance, the "Safety-First Model" implies that corporate insurers adjust their strategy primarily to:

[CO4]

A) Maximize short-term stock value
B) Minimize the probability of technical insolvency or ruin
C) Eliminate the cross-elasticity of alternative lines
D) Standardize products for weaker social groups

6. An unexpected fire that physically damages a commercial warehouse is classified as a: [CO1]

A) Peril
B) Moral Hazard
C) Morale Hazard
D) Speculative Risk

7. When a corporate risk manager charts an organization's exposures based on their estimated "frequency" and "severity," they are utilizing a: [CO6]
A) Product Development Funnel
B) Claim Development Triangle
C) Risk Management Matrix
D) Single Crossing Property Grid

8. An individual leaves their car unlocked with the keys inside simply because they know the vehicle is fully covered by comprehensive auto insurance. This behaviour is a direct example of: [CO6]
A) Adverse Selection
B) Symmetric Underwriting
C) Moral Hazard
D) Finite Risk Transfer

9. A major reinsurance technique where an insurance firm transfers predefined blocks of risk to the mainstream capital markets is an example of an: [CO6]
A) Alternative Risk Transfer (ART) instrument
B) Asymmetric Information Loop
C) Internal Line Departmentalization
D) Traditional Pooling Arrangement

10. In a classic historical Claim Development Triangle used to track changes over time, what do the diagonals typically represent? [CO5]
A) The exact accident years under analysis
B) The specific calendar years in which claims were actually paid or evaluated
C) The lag intervals or development periods elapsed
D) The ultimate terminal loss estimation bounds

(10x1=10)

Part B

Short Answer Questions (30 Words)

Answer any 10 questions

Each question carries 3 marks

11. Briefly outline the general scope and historic importance of Actuarial Science within modern financial markets. [CO1]
12. List four fundamental professional skills required for an actuary to perform effective liabilities valuation. [CO1]
13. Explain the core objective behind calculating the Human Life Value (HLV) in financial underwriting. [CO3]

14. Identify the primary purpose of the IRDAI Micro Insurance Regulations for weaker socio-economic sections. [CO1]
15. Differentiate briefly between price elasticity and cross-elasticity of demand for insurance products. [CO2]
16. What is the fundamental difference between a Group Gratuity scheme and a Group Superannuation scheme in India? [CO2]
17. Distinguish clearly between a 'Hazard' and a 'Peril' using a single contextual example. [CO1]
18. Outline two major contributions of a structured Risk Management Process to corporate operational continuity. [CO6]
19. Define the concept of 'Adverse Selection' as it occurs in an insurance market before a policy is issued. [CO6]
20. Summarize how a risk manager utilizes a basic risk management matrix to choose between risk avoidance and risk retention. [CO6]
21. What is meant by "Finite Risk Reinsurance" in the alternative risk financing landscape? [CO6]
22. Define 'Departmentalization' and state how it alters corporate efficiency in general insurance company operations. [CO5]
23. State the primary purpose of constructing a historical Claim Development Triangle in property and casualty insurance. [CO5]

(10x3=30)

Part C

Short Essay Questions (150 Words)

Answer any 5 questions

Each question carries 6 marks

24. Explain the multifaceted role of an actuary in a Life Insurance company, focusing specifically on product pricing, liability valuation, and assessment of systemic solvency. [CO1]
25. Analyze the purpose, scope, and operational objectives of financial underwriting. Explicitly demonstrate the core economic concepts of *insurable interest* and *value assessment* during personal insurance verification. [CO3]
26. Discuss the economic factors that heavily influence insurance consumption trends across developing nations [CO2]
27. Compare and contrast intuitive decision-making frameworks with strict mathematical models under asymmetric information. Address how insurers deploy safety-first strategies to navigate consumer moral hazards. [CO4]
28. Explain the systematic components of building an effective corporate Risk Management Programme. Highlight the key organizational steps, policy formulation guidelines, and explicit duties of a professional risk manager. [CO6]
29. Critique the expanding utility of the Alternative Risk Transfer (ART) market. Discuss how modern entities exploit capital markets as an additional resource for risk capacity and alternative financing. [CO6]
30. Outline the standard Product Development Process within modern insurance organizations. Using an illustrative mock matrix grid layout, explain the operational components (Rows, Columns, and Diagonals) of a basic Claim Development Triangle to track ultimate outstanding claims. [CO5]

(5x6=30)